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MATH2921 Tutorial Sheet 7 – Solution (Week 8)

Semester 1, 2025 | Vector Calculus and Differential Equations (Advanced)

Suggestion: try all problems, ideally before tutorial class. Feel free to use e-visual supports in the process, although it is ultimately important knowing how to proceed with just pen and paper.

1. The equation of motion for a damped harmonic oscillator is given by

$$m\ddot{y} + 2\delta\dot{y} + ky = 0$$

where $y(t)$ is the the oscillator's position relative to rest, m is the mass of the oscillator, 2δ is the damping coefficient and k is the constant of proportionality of the restoring force/spring.

(a) Write this second order constant coefficient ODE out as a first order linear system of 2 equations (i.e. in the form $\dot{\mathbf{x}} = A\mathbf{x}$).

(b) What is the characteristic equation of the matrix you found in part (a)? How does this compare to the characteristic equation of the ODE (that one you get when you plug in a solution of the form e^{rt})?

(c) Suppose that $\delta^2 - km > 0$. Write the general solution to the ODE using three different methods. First by finding the roots of the characteristic equation of the ODE, and the second by finding the eigenvalues and eigenvectors of the matrix you found in part (a), and thirdly by computing the matrix exponential of the matrix in part (a).

Solution:

(a) First we divide by m to rewrite the equation as

$$\ddot{y} + \frac{2\delta}{m}\dot{y} + \frac{k}{m}y = 0.$$

Then we set $x_1 \equiv y$ and $x_2 \equiv \dot{y}$ and we have that the ODE is equivalent to the first order linear system

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{2\delta}{m} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =: A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

(b) The characteristic equation of the matrix A is

$$\rho_A(\lambda) = \det \begin{vmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\frac{2\delta}{m} - \lambda \end{vmatrix} = \lambda^2 + \frac{2\delta}{m}\lambda + \frac{k}{m} = 0.$$

The characteristic equation of the ODE is

$$mr^2 + 2\delta r + k = 0.$$

So these two equations have the same roots.

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(c) Part (b) shows that these two equations have the same roots, so the eigenvalues of A are the same as the roots of the characteristic equation of the ODE:

$$r_\pm = \frac{-\delta \pm \sqrt{\delta^2 - km}}{m}.$$

Since $\delta^2 - 4km > 0$ these roots are real, so the general solution is

$$y(t) = c_1 e^{r_+ t} + c_2 e^{r_- t}$$

for arbitrary real constants $c_{1,2}$.

The eigenvectors of the matrix A corresponding to the eigenvalues $r_\pm$ are $\begin{pmatrix} 1 \\ r_\pm \end{pmatrix}$ respectively. Thus the general solution to the first order system of ODEs is

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ r_+ \end{pmatrix} e^{r_+ t} + c_2 \begin{pmatrix} 1 \\ r_- \end{pmatrix} e^{r_- t}$$

or in matrix form

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} e^{r_+ t} & e^{r_- t} \\ r_+ e^{r_+ t} & r_- e^{r_- t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

If $b^2 - 4ac > 0$, then $r_+ \neq r_-$ and so we have two distinct roots of the characteristic equation. Letting

$$P = \begin{pmatrix} 1 & 1 \\ r_- & r_+ \end{pmatrix}, \quad P^{-1} = \frac{1}{r_+ - r_-} \begin{pmatrix} r_+ & -1 \\ -r_- & 1 \end{pmatrix}, \quad \text{and } D = \begin{pmatrix} r_- & 0 \\ 0 & r_+ \end{pmatrix}$$

We have that

$$At = PDtP^{-1} \quad \text{and} \quad e^{At} = Pe^{Dt}P^{-1}$$

plugging values in and setting $\sqrt{\delta^2 - km} =: \Delta$, we have

$$e^{At} = e^{\frac{-t\delta}{m}} \begin{pmatrix} \cosh(\frac{t}{m}\Delta) + \frac{\delta \sinh(\frac{t}{m}\Delta)}{\Delta} & \frac{m \sinh(\frac{t}{m}\Delta)}{\Delta} \\ -\frac{k \sinh(\frac{t}{m}\Delta)}{\Delta} & \cosh(\frac{t}{m}\Delta) - \frac{\delta \sinh(\frac{t}{m}\Delta)}{\Delta} \end{pmatrix}$$

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2. Compute e^A and e^{At} for the following matrices (exactly):

(a) $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$ (c) $\begin{pmatrix} -1 & 2 \\ 4 & 3 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

(e) $\begin{pmatrix} 2 & -1 & -1 \\ -1 & 0 & -1 \\ 1 & 3 & 4 \end{pmatrix}$.

Solution:

(a) This is a diagonal matrix so $e^A = \begin{pmatrix} e^2 & 0 \\ 0 & e \end{pmatrix}$, $\begin{pmatrix} e^{2t} & 0 \\ 0 & e^t \end{pmatrix}$.

(b) The eigenvalues of the matrix are 1 and 2. With eigenvectors $v_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$, and $v_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ respectively. So we set $P = \begin{pmatrix} -3 & 1 \\ 1 & 0 \end{pmatrix}$, then $P^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$ for $D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. Finally $e^A = \begin{pmatrix} e^2 & 3\left(e^2 - e\right) \\ 0 & e \end{pmatrix}$, $e^{At} = \begin{pmatrix} e^{2t} & 3e^t\left(-1 + e^t\right) \\ 0 & e^t \end{pmatrix}$

(c) We have

$$P = \begin{pmatrix} \frac{1}{2}\left(-1 + \sqrt{3}\right) & \frac{1}{2}\left(-1 - \sqrt{3}\right) \\ 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{6}\left(3 + \sqrt{3}\right) \\ -\frac{1}{\sqrt{3}} & \frac{1}{6}\left(3 - \sqrt{3}\right) \end{pmatrix} \quad \text{and}$$

$$D = \begin{pmatrix} 1 + 2\sqrt{3} & 0 \\ 0 & 1 - 2\sqrt{3} \end{pmatrix}$$

so

$$e^A = \begin{pmatrix} e\left(\cosh\left(2\sqrt{3}\right) - \frac{\sinh\left(2\sqrt{3}\right)}{\sqrt{3}}\right) & \frac{e\sinh\left(2\sqrt{3}\right)}{\sqrt{3}} \\ \frac{2e\sinh\left(2\sqrt{3}\right)}{\sqrt{3}} & e\left(\cosh\left(2\sqrt{3}\right) + \frac{\sinh\left(2\sqrt{3}\right)}{\sqrt{3}}\right) \end{pmatrix}$$

and $e^{At} = \begin{pmatrix} e^t\cosh\left(2\sqrt{3}t\right) - \frac{e^t\sinh\left(2\sqrt{3}t\right)}{\sqrt{3}} & \frac{e^t\sinh\left(2\sqrt{3}t\right)}{\sqrt{3}} \\ \frac{2e^t\sinh\left(2\sqrt{3}t\right)}{\sqrt{3}} & e^t\cosh\left(2\sqrt{3}t\right) + \frac{e^t\sinh\left(2\sqrt{3}t\right)}{\sqrt{3}} \end{pmatrix}$.

(d) We have

$$P = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad \text{and} \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$$

so

$$e^A = \begin{pmatrix} \frac{1}{2}\left(1 + e^2\right) & \frac{1}{2}\left(1 - e^2\right) \\ \frac{1}{2}\left(1 - e^2\right) & \frac{1}{2}\left(1 + e^2\right) \end{pmatrix}, \quad \text{and} \quad e^{At} = \begin{pmatrix} e^t\cosh(t) & -e^t\sinh(t) \\ -e^t\sinh(t) & e^t\cosh(t) \end{pmatrix}$$

(e) We have

$$P = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ -1 & -2 & -1 \end{pmatrix}$$

and

$$D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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so,

$$e^A = \begin{pmatrix} e^2 & -e^3+e^2 & -e^3+e^2 \\ -e^2+e & -e^2+2e & -e^2+e \\ e^2-e & e^3+e^2-2e & e^3+e^2-e \end{pmatrix},$$

and

$$e^{At} = \begin{pmatrix} e^{2t} & -e^{3t}+e^{2t} & -e^{3t}+e^{2t} \\ -e^{2t}+e^t & -e^{2t}+2e^t & -e^{2t}+e^t \\ e^{2t}-e^t & e^{3t}+e^{2t}-2e^t & e^{3t}+e^{2t}-e^t \end{pmatrix}.$$

Remark: it's mostly just the standard diagonalization process. For a matrix A, the Characteristic Equation is

$$\det(A-\lambda I)=0$$

and we solve for λ to get the eigenvalues. For each eigenvalue λ, we solve the algebraic linear homogeneous system

$$(A-\lambda I)\mathbf{v}=\underline{0}$$

to get he eigenvectors. Then we list the eigenvectors as column to get P and the corresponding diagonal matrix D for the diagonalization

$$A = PDP^{-1}.$$

Then by the features of Taylor series used for the likes of e^A, $\sin A$, $\cos A$, we have

$$e^A = P(e^D)P^{-1}, \sin A = P(\sin D)P^{-1}, \cos A = P(\cos D)P^{-1}$$

where e^D, $\sin D$ and $\cos D$ are just the diagonal matriices with diagonal entries being e^λ, $\sin\lambda$ and $\cos\lambda$ respectively, still just by the Taylor series used for these matrices.

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3. Consider the function:

$$H(x,y) = \frac{1}{2}(y^2 - x^2) + xy$$

Let's suppose that x and y are actually functions of t. Further suppose that

$$\begin{aligned} \dot{x} &= \frac{dx}{dt} = \frac{\partial H}{\partial y} \qquad \text{and,} \\ \dot{y} &= \frac{dy}{dt} = -\frac{\partial H}{\partial x}. \end{aligned} \tag{1}$$

Such a system is called a *Hamiltonian system*, and the function $H(x,y)$ is called the *Hamiltonian* of (1).

(a) Write out (1) as a 2×2 linear autonomous ODE. Write the general solution to it.

(b) Show that $\frac{dH}{dt} = 0$. This means that, in particular, the *level sets* of H are constant in t.

(c) Plot the level sets of H in terms of the variables x and y. Show that the vector $(\dot{x}, \dot{y})^\top$ is tangent to the level sets.

(d) You can visualise the solution to (1) by taking your initial condition, and finding which level set in the (x,y) plane it lies on. Then moving it according to the direction of the vector $(\dot{x}, \dot{y})$. Do this for a couple of initial conditions. These are called *phase curves* of (1).

(e) Choose an initial condition on an eigenvector (there's two, so you should do this for both) of A, what level set of H does it lie on? Plot the phase curve corresponding to such an initial condition.

Solution:

(a)

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

The eigenvalues are $\lambda_\pm = \pm\sqrt{2}$ with corresponding eigenvectors $v_\pm = \begin{pmatrix} 1 \pm \sqrt{2} \\ 1 \end{pmatrix}$.

Therefore two linearly independent solutions are given by $x_\pm(t) = e^{\pm\sqrt{2}t} \begin{pmatrix} 1 \pm \sqrt{2} \\ 1 \end{pmatrix}$, and the general solution is

$$c_1 x_+(t) + c_2 x_-(t) = c_1 e^{\sqrt{2}t} \begin{pmatrix} 1 + \sqrt{2} \\ 1 \end{pmatrix} + c_2 e^{-\sqrt{2}t} \begin{pmatrix} 1 - \sqrt{2} \\ 1 \end{pmatrix}$$

where $c_{1,2}$ are arbitrary constants. Alternatively, you can write the general solution as

$$e^{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} t} x_0$$

where

$$e^{At} = \begin{pmatrix} \cosh\left(\sqrt{2}t\right) + \frac{\sinh\left(\sqrt{2}t\right)}{\sqrt{2}} & \frac{\sinh\left(\sqrt{2}t\right)}{\sqrt{2}} \\ \frac{\sinh\left(\sqrt{2}t\right)}{\sqrt{2}} & \cosh\left(\sqrt{2}t\right) - \frac{\sinh\left(\sqrt{2}t\right)}{\sqrt{2}} \end{pmatrix}$$

(b)

$$\frac{dH}{dt} = H_x \frac{dx}{dt} + H_y \frac{dy}{dt} = H_x H_y - H_y H_x = 0$$

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(c) & (d) One can try to plot it in $\mathbb{R}^2$ which can be done by online tools. It's a standard parabola with a simple linear transform. Part (b) shows that the gradient of H is perpendicular to $(\dot{x}, \dot{y})^\top$, and so the vector must be tangent to the level sets of H.

(d) $H(k(1+\sqrt{2}), k) = H(k(1-\sqrt{2}), k) = 0$ for an arbitrary $k \in \mathbb{R}$. So the eigenvectors all lie on the level sets $H = 0$. These are the two straight lines in the left box. The slopes of the lines correspond to the $v_\pm$. For the phase curve, it is clear that the phase curve starting on v_- is going into the origin, and the phase curve for v_+ is going away from it.

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4. Prove that for $n \times n$ matrices A and B, $e^{At}e^{Bt} = e^{(A+B)t}$ for all real numbers t if and only if $AB - BA = [A, B] = 0$.

Solution:

$\Leftarrow$ If $AB - BA = 0 \Rightarrow AtBt - BtAt = 0$ for all $t \in \mathbb{R}$. This implies $e^{At}e^{Bt} = e^{(A+B)t}$ for all real t (from class).
$\Rightarrow$ If $e^{At}e^{Bt} = e^{(A+B)t}$ for all real t, then differentiating both sides with respect to t gives

$$\begin{aligned}
Ae^{At}e^{Bt} + e^{At}Be^{Bt} &= (A+B)e^{(A+B)t} && \Rightarrow \\
Ae^{At}e^{Bt} + e^{At}Be^{Bt} &= Ae^{(A+B)t} + Be^{(A+B)t} && \Rightarrow \\
e^{At}Be^{Bt} &= Be^{At}e^{Bt} && \Rightarrow \\
e^{At}B &= Be^{At} \qquad \forall t \in \mathbb{R}.
\end{aligned}$$

So, e^{At} and B commute for all $t \in \mathbb{R}$. Differentiating a second time gives

$$\begin{aligned}
Ae^{At}B &= BAe^{At} \Rightarrow \quad \big(\text{because } B \text{ and } e^{At} \text{ commute}\big) \\
ABe^{At} &= BAe^{At} \Rightarrow \quad \big(\text{because } e^{At} \text{ is invertible}\big) \\
AB &= BA.
\end{aligned}$$

$\square$

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5. The *algebraic multiplicity* of an eigenvalue λ of an $n \times n$ real matrix is defined as the multiplicity of the eigenvalue as a root of the characteristic equation. That is if λ is the eigenvalue, and you can write

$$\det(A - x\mathbb{I}) = (x - \lambda)^a \rho(x)$$

for some positive integer a, where $\rho(\lambda) \neq 0$, then we say that λ has algebraic multiplicity a. The *geometric multiplicity* of an eigenvalue is defined to be the dimension of the eigenspace. Show that the geometric multiplicity of an eigenvalue is always less than or equal to its algebraic multiplicity.

SOLUTION:

Let A be the $n \times n$ matrix of a linear transformation, written in the standard basis. Suppose that λ is an eigenvalue with geometric multiplicity k. This means that there are k linearly independent eigenvectors $\mathbf{v}_1, \dots \mathbf{v}_k$, all with the same eigenvalue. Writing A out in the basis $\{\mathbf{v}_1, \dots \mathbf{v}_k, \mathbf{e}_{k+1}, \dots, \mathbf{e}_n\}$, we have $A = \begin{pmatrix} \lambda\mathbb{I} & B \\ 0 & \tilde{A} \end{pmatrix}$, where $\mathbb{I}$ is the $k \times k$ identity matrix and $\begin{pmatrix} B \\ \tilde{A} \end{pmatrix}$ is the matrix whose columns are $A\mathbf{e}_j$, $j = k+1, \dots n$. Now the characteristic polynomial $\rho_A(x)$ is independent of basis. (To see this observe that $\det(A - x\mathbb{I}) = \det(P^{-1}(A - x\mathbb{I})P) = \det(P^{-1}AP - x\mathbb{I})$.) This means that the characteristic polynomial of A, $\rho_A(x) = (\lambda - x)^k \rho_{\tilde{A}}(x)$. This means that the algebraic multiplicity of the eigenvalue λ is at least k.

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6. Find all possible values of a, b, c, d for which the 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is diagonalisable.

Solution:

The matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ will be diagonalisable provided it either has 1) two distinct eigenvalues, or 2) has equal eigenvalues but two eigenvectors. The eigenvalues are

$$\lambda_{\pm} = \frac{a + d \pm \sqrt{(a-d)^2 + 4bc}}{2}.$$

So if we have that any a, b, c, d such that $(a-d)^2 \neq -4bc$, A is diagonalisable with 2 distinct eigenvalues.
If A has a repeated eigenvalue λ but is still diagonalisable, it must be a diagonal matrix.

So we need either $(a-d)^2 \neq -4bc$ or $a = d$ and $b = c = 0$ for A to be diagonalisable.

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7. Prove that if A and B are *similar* matrices (i.e. $B = P^{-1}AP$ for some invertible matrix P), then they have the same eigenvalues and that these have the same multiplicities.

Solution:

A and B will have the same characteristic polynomial.

$$\begin{aligned}
\det(B - \lambda\mathbb{I}) &= \det(P^{-1}AP - \lambda\mathbb{I}) \\
&= \det(P^{-1}(A - \lambda\mathbb{I})P) \\
&= \det(P^{-1})\det(A - \lambda\mathbb{I})\det(P) \\
&= \det(A - \lambda\mathbb{I}).
\end{aligned}$$

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8. In this problem we're going to explore some more functions of matrices. For a real number t, we have two different, but equivalent ways to define the functions $\sin(t)$ and $\cos(t)$, using either the Taylor series, or Euler's formula. That is:

$$\sin(t) = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!} \quad \text{or,} \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}.$$

Likewise we have:

$$\cos(t) = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!} \quad \text{or,} \quad \cos(t) = \frac{e^{it} + e^{-it}}{2}.$$

It makes sense for us to do the same thing with matrices, either using the power series, or the exponential of a matrix. That is we can *define* the sine or cosine of a matrix using either of the above definitions. We have for *any* $n \times n$ matrix A:

$$\sin(A) := \frac{e^{Ai} - e^{-Ai}}{2i} \quad \text{or,} \quad \sin(A) := \sum_{k=0}^{\infty} \frac{(-1)^k A^{2k+1}}{(2k+1)!}$$

and similarly for:

$$\cos(A) := \frac{e^{Ai} + e^{-Ai}}{2} \quad \text{or,} \quad \cos(A) := \sum_{k=0}^{\infty} \frac{(-1)^k A^{2k}}{(2k)!}.$$

(a) Pick either $\sin(A)$ or $\cos(A)$ and show that the two definitions are equivalent.

(b) Compute $\sin(A)$ and $\cos(A)$ for the matrices

i. $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

ii. $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

iii. $A = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 0 & -1 \\ 1 & 3 & 4 \end{pmatrix}$.

(c) Show that (as matrices now), $\sin^2(A) + \cos^2(A) = \mathbb{I}$ for any $n \times n$ matrix A.

Just as we did for the exponential, for a fixed $n \times n$ matrix A, we can now define *functions* $\sin(At) : \mathbb{R} \to M_n(\mathbb{R})$ and $\cos(At) : \mathbb{R} \to M_n(\mathbb{R})$.

(d) Compute $\sin(At)$ and $\cos(At)$ for the matrices in part (b).

(e) Show that $\frac{d}{dt}\sin(At) = A\cos(At)$, and that $\frac{d}{dt}\cos(At) = -A\sin(At)$.

(f) Suppose that A is an invertible $n \times n$ matrix. Show that the initial value problem $\ddot{\mathbf{x}} = -A^2\mathbf{x}$, with $\mathbf{x}(0) = 0$ and $\dot{\mathbf{x}}(0) = x_0$ is solved by $A^{-1}\sin(At)x_0$. Show that the initial value problem $\ddot{\mathbf{x}} = -A^2\mathbf{x}$, with $\mathbf{x}(0) = x_0$ and $\dot{\mathbf{x}}(0) = 0$ is solved by $\cos(At)x_0$

(g) Suppose that A is an invertible matrix. Write $\ddot{\mathbf{x}} = -A^2\mathbf{x}$ as a $2n \times 2n$ first order system of equations. Find the general solution. (Hint: use the transformation $\dot{\mathbf{x}} = -A\mathbf{y}$ to put the system in a 'familiar' form). What happens when A is not invertible?

Solution:

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(a) I chose cosine. You should do this exercise for sine yourself if you need to.

$$\frac{e^{Ai}+e^{-Ai}}{2} =$$
$$\frac{1}{2}\left[\left(\mathbb{I}+Ai-\frac{1}{2}A^2-\frac{1}{6}A^3i+\frac{1}{4!}A^4+\cdots+\right)+\left(\mathbb{I}+Ai-\frac{1}{2}A^2-\frac{1}{6}A^3i+\frac{1}{4!}A^4+\cdots\right)\right]$$
$$=2\left(\frac{1}{2}\right)\left(\mathbb{I}-\frac{1}{2}A^2+\frac{1}{4!}A^4-\cdots+\right)=\cos(A).$$

(b) See part (d), and plug in $t=1$.

(c)

$$\begin{aligned}\sin^2(A)+\cos^2(A) &= -\frac{1}{4}\left(e^{Ai}-e^{-Ai}\right)^2+\frac{1}{4}\left(e^{Ai}+e^{-Ai}\right)^2\\ &= -\frac{1}{4}\left(e^{2Ai}-2\mathbb{I}+e^{-2Ai}\right)+\frac{1}{4}\left(e^{2Ai}+2\mathbb{I}+e^{-2Ai}\right)\\ &= \frac{1}{4}\left(2\mathbb{I}+2\mathbb{I}\right)=\mathbb{I}\end{aligned}$$

(d) i. Since this is a diagonal matrix we have that

$$\cos(At)=\begin{pmatrix}\cos(2t) & 0\\ 0 & \cos(t)\end{pmatrix} \quad \text{and} \quad \sin(At)=\begin{pmatrix}\sin(2t) & 0\\ 0 & \sin(t)\end{pmatrix}$$

ii. Using diagonalisation with complex eigenvalues, we have that

$$e^{At}=e^t\begin{pmatrix}\cos t & -\sin t\\ \sin t & \cos t\end{pmatrix}$$

Using this and the exponential formulas that say $\sin(it)=i\sinh(t)$ and $\cos(it)=\cosh(t)$ we have

$$\cos(At)=\begin{pmatrix}\cos(t)\cosh(t) & \sin(t)\sinh(t)\\ -\sin(t)\sinh(t) & \cos(t)\cosh(t)\end{pmatrix}$$

and

$$\sin(At)=\begin{pmatrix}\sin(t)\cosh(t) & -\cos(t)\sinh(t)\\ -\cos(t)\sinh(t) & \sin(t)\cosh(t)\end{pmatrix}$$

iii. For this matrix, it's not hard to get the Characteristic equation into the form

$$(\lambda-1)(\lambda-2)(\lambda-3)=0$$

for eigenvalues 1, 2 and 3. One can then find the eigenvectors for diagonalization. The calculation would take some time, but it's most important to know how to do this by hand.
We have (after Mathematica FullSimplify)

$$e^{At}=\begin{pmatrix}e^{2t} & -e^{2t}\left(-1+e^t\right) & -e^{2t}\left(-1+e^t\right)\\ -e^t\left(-1+e^t\right) & -e^t\left(-2+e^t\right) & -e^t\left(-1+e^t\right)\\ e^t\left(-1+e^t\right) & e^t\left(-2+e^t+e^{2t}\right) & e^t\left(-1+e^t+e^{2t}\right)\end{pmatrix}$$

so

$$\cos(At)=\begin{pmatrix}\cos(2t) & \cos(2t)-\cos(3t) & \cos(2t)-\cos(3t)\\ \cos(t)-\cos(2t) & 2\cos(t)-\cos(2t) & \cos(t)-\cos(2t)\\ \cos(2t)-\cos(t) & -2(6\cos(t)+2\cos(2t)+3)\sin^2\left(\frac{t}{2}\right) & -\cos(t)+\cos(2t)+\cos(3t)\end{pmatrix}$$

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and

$$\sin(At) = \begin{pmatrix} \sin(2t) & \sin(2t) - \sin(3t) & \sin(2t) - \sin(3t) \\ \sin(t) - \sin(2t) & 4\sin^2\left(\frac{t}{2}\right)\sin(t) & \sin(t) - \sin(2t) \\ (2\cos(t) - 1)\sin(t) & -2\sin(t) + \sin(2t) + \sin(3t) & 8\cos^3\left(\frac{t}{2}\right)\left(\sin\left(\frac{3t}{2}\right) - 2\sin\left(\frac{t}{2}\right)\right) \end{pmatrix}$$

These were both computed with the aid of Mathematica.

(e) We prove this by direct computation:

$$\frac{d}{dt}\sin(At) = \frac{d}{dt}\left(\frac{e^{Ait} - e^{-Ait}}{2i}\right) = \frac{Aie^{Ait} + Aie^{-Ait}}{2i} = A\cos(At)$$
$$\frac{d}{dt}\cos(At) = \frac{d}{dt}\left(\frac{e^{Ait} + e^{-Ait}}{2}\right) = \frac{Aie^{Ait} - Aie^{-Ait}}{2} = -A\sin(At)$$

(f) Let $x(t) = A^{-1}\sin(At)x_0$. Then we have that $x(0) = 0$ and

$$\begin{aligned} \dot{x}(t) &= \cos(At)x_0 \\ \dot{x}(0) &= \mathbb{I} \\ \ddot{x}(t) &= -A\sin(At)x_0 = -A^2x(t) \end{aligned}$$

Now let $x(t) = \cos(At)x_0$. Then we have that $x(0) = x_0$ and

$$\begin{aligned} \dot{x}(t) &= -A\sin(At)x_0 \\ \dot{x}(0) &= 0 \\ \ddot{x}(t) &= -A^2\cos(At)x_0 = -A^2x(t) \end{aligned} \tag{2}$$

(g) If $\dot{x} = -Ay$, then we have that $y = -A^{-1}\dot{x}$ and so $\dot{y} = -A^{-1}\ddot{x} = Ax$. We write this as a $2n \times 2n$ first order system

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 0 & -A \\ A & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

The general solution is therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = e^{\begin{pmatrix} 0 & -A \\ A & 0 \end{pmatrix}t}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \cos(At) & -\sin(At) \\ \sin(At) & \cos(At) \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \quad \text{(a } 2n \times 2n \text{ system)}$$

When A is not invertible, you can't make the transformation $-A^{-1}\dot{x} = y$. Instead write $\dot{x} = y$. Then you have a $2n \times 2n$ system

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 0 & \mathbb{I} \\ -A^2 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

Again, the general solution is

$$e^{\begin{pmatrix} 0 & \mathbb{I} \\ -A^2 & 0 \end{pmatrix}t}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix},$$

only this time $e^{\begin{pmatrix} 0 & \mathbb{I} \\ -A^2 & 0 \end{pmatrix}t}$ doesn't have a particularly nice form. You can write it a bit like how you would expect (as if A were a scalar), except the top right entry

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no longer works...

$$e^{\begin{pmatrix} 0 & \mathbb{I} \\ -A^2 & 0 \end{pmatrix} t} = \begin{pmatrix} \cos(At) & f(At) \\ -A\sin(At) & \cos(At) \end{pmatrix}$$

where

$$f(At) = t\mathbb{I} - \frac{1}{3!}A^2t^3 + \frac{1}{5!}A^4t^5 + \cdots$$

which would be $A^{-1}\sin(At)$ but A^{-1} doesn't exist, so we just have that $f(At)$ is just defined through the (evidently convergent) power series.

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